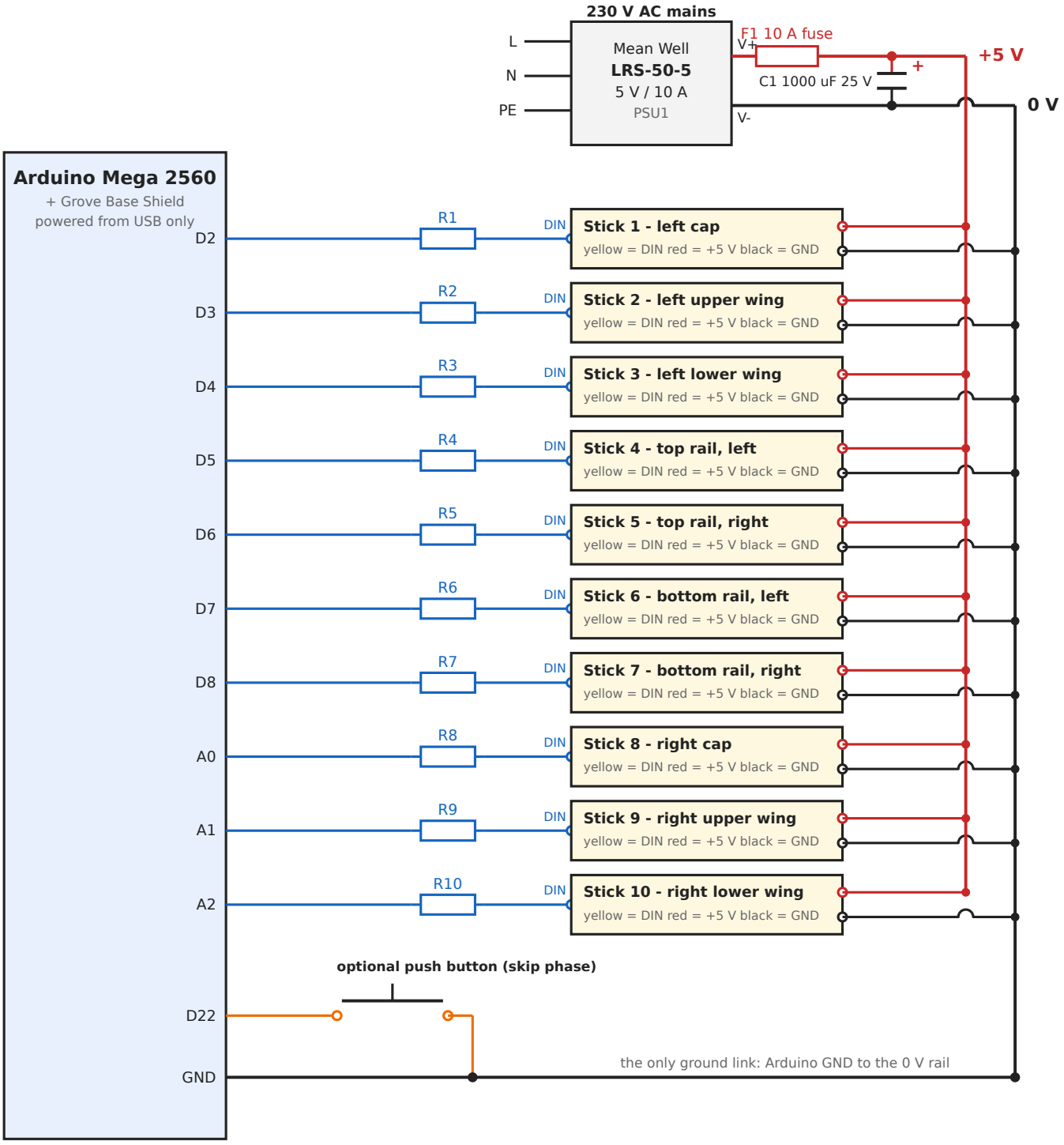
Stick test - wiring (Arduino Mega + 10 Grove RGB LED sticks on a separate 5 V supply)



Connections

1. The sticks are powered from the separate 5 V supply PSU1 through the 10 A fuse F1. The Arduino 5V pin does NOT power the sticks.
2. Each stick: red wire = +5 V rail, black wire = 0 V rail (thick wire for the rails). Only the yellow wire goes to the Arduino.
3. Take the red and black wire terminals out of the shield-side Grove connector of every stick cable (see README.txt), so that the shield and the stick supply are not connected to each other through the cables.
4. Data: Arduino pin -> 330 ohm resistor (recommended) -> yellow wire (DIN). Stick n goes to the n-th pin: D2..D8, A0..A2.
5. The Arduino and the 5 V supply share ground through ONE wire: Arduino GND to the 0 V rail.
6. C1: 1000 uF, 25 V electrolytic capacitor across the rails (plus = +5 V, stripe = minus), close to the sticks.
7. Switch the 5 V supply ON first, then plug in the USB cable. The USB cable powers only the Arduino.
8. Optional push button between D22 and GND: skips to the next phase.
9. The mains side of the supply must be in an enclosure and wired by a person who knows how (nothing here is mains).